Marketing Campaign & Product Performance – Maven Marketing

Short Description

A focused analysis of customer-level data for Maven Marketing to understand which campaigns most successful and which product categories are generate the highest revenue. The work supports data-driven decisions on where to invest future marketing budget and product focus.

Business Questions

(https://mavenanalytics.io/data-playground/marketing-campaign-results)

1. Which marketing campaign is most successful in terms of customer acceptance?
2. Which product categories are currently performing best in terms of revenue?

Approach

I worked with a customer-level dataset containing 2,240 records and variables on demographics, campaign responses and spending across multiple product categories over the last two years. After cleaning column names and checking for missing values, I confirmed that the key campaign response fields and product spend variables were fully complete, with only 24 missing values in income, which was not used in this analysis.

	variable	n_missing
5	income	24
1	id	0
2	year_birth	0
3	education	0
4	marital_status	0
6	kidhome	0
7	teenhome	0
8	dt_customer	0
9	recency	0
10	mnt_wines	0
11	mnt_fruits	0
12	mnt_meat_products	0
13	mnt_fish_products	0
14	mnt_sweet_products	0
15	mnt_gold_prods	0
16	num_deals_purchases	0
17	num_web_purchases	0
18	num_catalog_purchases	0
19	num_store_purchases	0
20	num_web_visits_month	0
21	accepted_cmp3	0
22	accepted_cmp4	0
23	accepted_cmp5	0
24	accepted_cmp1	0
25	accepted_cmp2	0
26	response	0
27	complain	0
28	country	0

To assess campaign performance, I calculated acceptance rates for each of the five historical campaigns and the most recent campaign by dividing the number of customers who accepted the offer (flag = 1) by the total number of customers. For product performance, I summed the total expenditure across all customers for wines, fruits, meat, fish, sweets and gold products and computed the average spend per customer in each category.

Key Findings

- **Campaign performance:**

The last campaign achieved an acceptance rate of approximately 14.9%, which is roughly double the rates observed for campaigns 3, 4 and 5 (around 7–7.5%) and far above campaign 2, which had an acceptance rate of only about 1.3%. This indicates that the most recent campaign was substantially more effective than previous efforts in converting customers.

```
> print(campaign_results)
        campaign acceptance_rate acceptance_pct
1     Campaign 1      0.06428571           6.43
2     Campaign 2      0.01339286           1.34
3     Campaign 3      0.07276786           7.28
4     Campaign 4      0.07455357           7.46
5     Campaign 5      0.07276786           7.28
6 Last Campaign      0.14910714          14.91
```

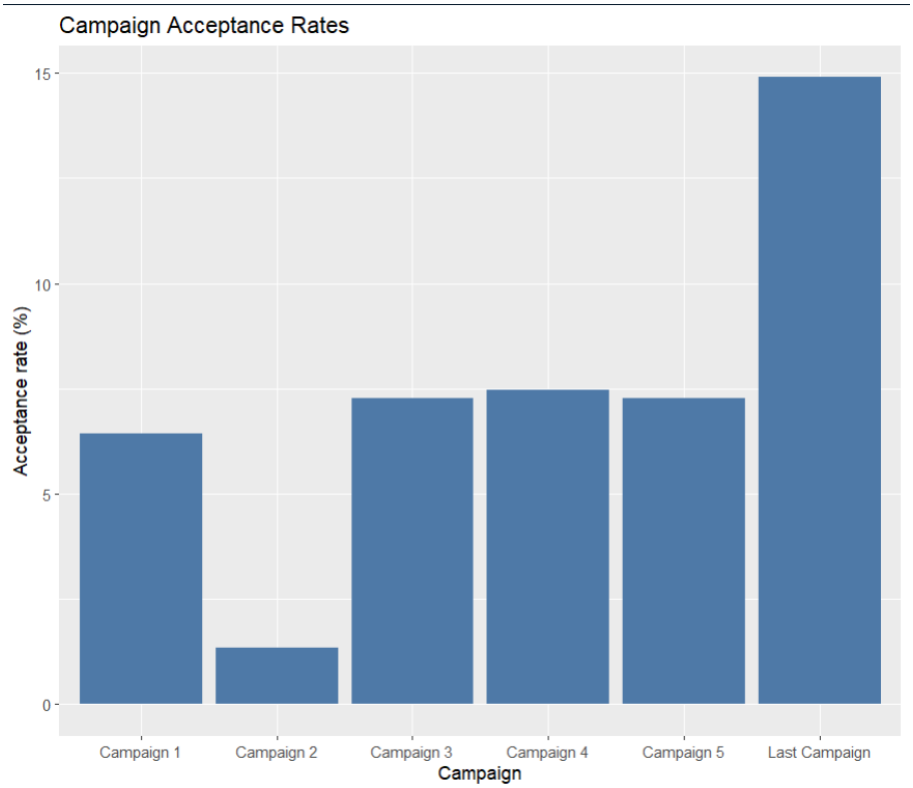


Figure 1: Campaign acceptance rates: the last campaign significantly outperforms all previous campaigns.

- **Product performance:**

Wines are the strongest-performing product category, with total customer spending of roughly 680,816 and an average of about 304 per customer. Meat products form the second largest revenue stream, with total spending around 373,968 and an average of about 167 per customer. Fruits and sweet products show considerably lower total and per-customer spend, suggesting they contribute less to overall revenue within this customer base.

```
> print(product_results)
          product total_spend avg_spend_per_customer
1           Wines      680816                 303.94
2          Fruits       58917                  26.30
3  Meat products      373968                 166.95
4  Fish products       84057                  37.53
5 Sweet products       60621                  27.06
6  Gold products       98609                  44.02
```

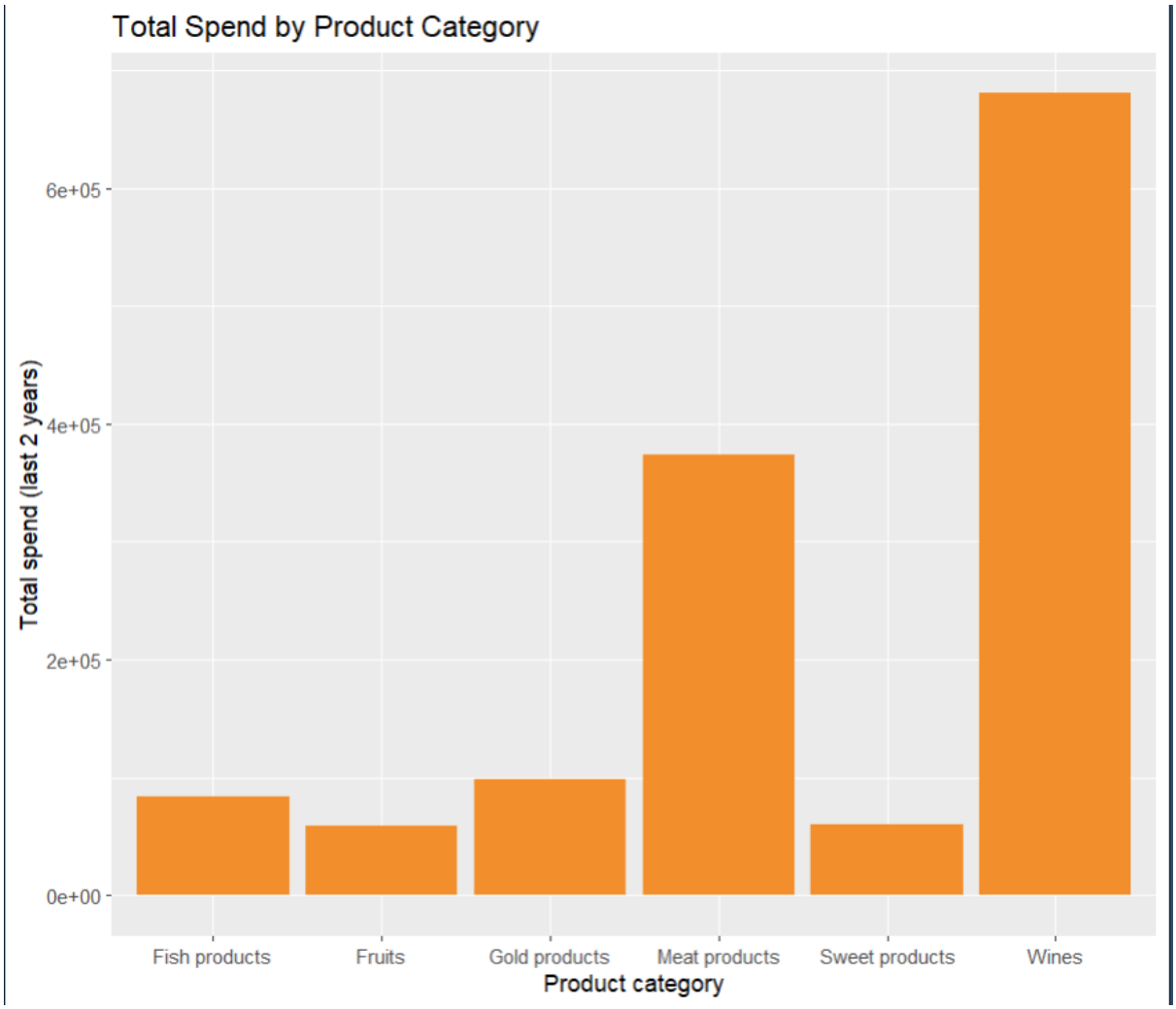


Figure 2: Total spend by product category over the last two years, showing wines and meat products as the dominant revenue drivers.

Recommendations

- Maven Marketing should investigate what differentiated the last campaign (offer design, targeting, timing or channels) and consider using or A/B testing those elements in future campaigns, rather than replicating the weaker set-ups seen in earlier campaigns, especially campaign 2.
- Given the strong performance of wines and meat products, future campaigns could prioritize these categories for promotions, bundles or loyalty offers, while exploring whether fruits and sweets can be repositioned (for example via cross-selling) to improve their contribution.